

Git and GitHub interview questions

Basic Git Questions

1. What is Git? How does it differ from other version control systems?
 2. What are the main features of Git?
 3. Explain the difference between Git and SVN.
 4. What are the advantages of using Git?
 5. What is the purpose of the `.git` directory in a Git repository?
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Git Commands

6. What is the purpose of the `git init` command?
 7. How do you create a new repository in Git?
 8. What is the difference between `git clone` and `git fork`?
 9. Explain the difference between `git pull` and `git fetch`.
 10. How do you check the status of your repository in Git?
 11. What is the command to stage files for a commit?
 12. How do you revert a commit that has already been pushed and made public?
 13. What does the `git reset` command do? Explain its options `--soft`, `--mixed`, and `--hard`.
 14. How do you squash commits in Git?
 15. What is the difference between `git merge` and `git rebase`?
 16. What is `git stash`, and how do you use it?
 17. How can you see the history of commits in Git?
 18. What is the difference between `git log` and `git reflog`?
 19. How do you tag a commit in Git?
 20. Explain the purpose of `git cherry-pick`.
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Branching and Merging

21. What are Git branches? Why are they used?
22. How do you create a new branch in Git?
23. How do you delete a branch in Git?
24. What is a merge conflict, and how do you resolve it?
25. How do you handle multiple branches in Git?
26. Explain fast-forward and three-way merge.
27. What is the purpose of `git branch -d` vs. `git branch -D`?
28. How do you rename a branch in Git?

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GitHub Questions

29. What is GitHub, and how does it relate to Git?
 30. What are repositories in GitHub?
 31. What is a pull request in GitHub?
 32. Explain the GitHub flow workflow.
 33. What is the difference between GitHub Actions and GitHub Workflows?
 34. How do you fork a repository on GitHub?
 35. What is a GitHub issue, and how can you use it?
 36. What are GitHub Projects?
 37. How can you collaborate with others on GitHub?
 38. What are webhooks in GitHub?
 39. Explain GitHub Pages and its uses.
 40. How do you secure your repository on GitHub?
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Advanced Git Concepts

41. What is a bare repository in Git?
 42. What is the **HEAD** in Git?
 43. What is the difference between **origin** and **upstream** in Git?
 44. Explain submodules in Git.
 45. How does Git handle large binary files?
 46. What are Git hooks? How are they used?
 47. What is Git LFS (Large File Storage)?
 48. How do you revert a file to a previous commit in Git?
 49. Explain the concept of a detached HEAD in Git.
 50. What is the difference between **git archive** and **git bundle**?
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Scenario-Based Questions

51. How would you handle a situation where you accidentally committed sensitive information to a repository?
52. What would you do if you want to ignore certain files in Git?
53. How can you ensure your local repository is in sync with the remote repository?
54. How would you resolve a situation where your branch is outdated compared to the main branch?
55. How do you recover a branch you deleted accidentally?

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Best Practices

- 56. What are some Git best practices you follow?
- 57. How do you maintain commit message standards?
- 58. What is the significance of `.gitignore`, and how do you configure it?
- 59. How do you ensure code quality and maintainability while collaborating with a team?
- 60. Explain the importance of version control in CI/CD pipelines.